Adopted Levels:not observed

S(n)=24210 *syst*; S(p)=−60 *syst*; Q(α)=−9610 *syst* 2021Wa16
ΔS(n)=500, ΔS(p)=360, ΔQ(α)=350 (syst,2021Wa16).
S(2p)=−2510 *300*, Q(ε)=16110 *360*, Q(εp)=16990 *300* (syst,2021Wa16).
Isotope discovery: none.
S(2p) is negative and S(p) may be consistent with zero. ^{34}Ca is a candidate for ground-state two-proton emitter.
2026Li03: S(p)=401 and S(2p)=−1987 from an average of Finite-Range Droplet Model (FRDM12), Koura-Tachibana-Uno-Yamada (KTUY) model, Weizsacker-Skyrme (WS4) model, and Duflo-Zuker (DZ31) model combined with the radial basis function mirror symmetry (RBFms) approach.
2024Li59: S(2p)=−1044, $T_{1/2}$=177.8 ns. The Q(2p) decay energy was determined using an empirical linear relationship. The $T_{1/2}$ was calculated by inputting the refined decay energy into the new Geiger-Nuttall formula.
2024Ji17: $T_{1/2}$=8.1 fs from an extension of the Bayrak model with and without the deformation term. Used Q(2p)=2510 from 2021Wa16.
2023Me08: $T_{1/2}$=64.6 ps. The WKB approximation is applied to a Yukawa-plus-exponential model potential. Used Q(2p)=1470 from 2017Wa10 and a spectroscopic factor derived from a specific cubic-polynomial relation.
2022Zo01: S(p)=259 *96*, S(2p)=−2132 *151*, mass excess=14497 *98*. Mass relations of mirror nuclei.
2022Sa41: $T_{1/2}$=2.2 fs from the Coulomb and proximity potential model for deformed nuclei (CPPMDN) incorporating ground-state deformation (β2) and orientation effect. Used Q(2p)=2510 from 2021Wa16.
2022Ro15: $T_{1/2}$=0.35 fs from the generalized liquid drop model (GLDM). $T_{1/2}$=3.0 as from a semi-empirical analytic expression based on a modified version of the Geiger-Nuttall law. Both calculations used Q(2p)=2510 from 2021Wa16.
2022Pa24: $T_{1/2}$=0.31 ns from a two-potential approach; the nuclear potential is obtained using the Skyrme-Hartree-Fock approach and the Skyrme effective interaction of SLy8. Used Q(2p)=1470 from 2017Wa10.
2021Xi06: $T_{1/2}$=3.5 fs from the Unified Fission Model (UFM), $T_{1/2}$=1.7 fs from the Generalized Liquid Drop Model (GLDM), $T_{1/2}$=27.5 fs from the Effective Liquid Drop Model (ELDM). These models share a similar physical mechanism by treating the decay as a ^{2}He cluster preformed on the nuclear surface that subsequently penetrates the Coulomb barrier. All used Q(2p)=2510 from 2021Wa16.
2021Li27: $T_{1/2}$=79.4 ps from the Gamow-like model. Used Q(2p)=1470 from 2017Wa10.
2021Li16: $T_{1/2}$=117 ps from a two-parameter empirical formula based on the Geiger-Nuttall law. Used Q(2p)=1470 from 2017Wa10.
2020Ma35: S(p)=203 *151*, S(2p)=−2152 *118*, $T_{1/2}$=3.6 ps *+51−20*. Mass relations of mirror nuclei with local correlations and a four-parameter empirical formula based on the Effective Liquid Drop Model (ELDM).
2020Cu01: $T_{1/2}$=19.5 ps from a generalized liquid drop model (GLDM). Used Q(2p)=1470 from 2017Wa10.
2019Sr02: $T_{1/2}$=2.2 ns. A four-parameter empirical formula was fitted using the theoretical results of the Effective Liquid Drop Model (ELDM). Used Q(2p)=1470 from 2017Wa10.
Other decay half-life calculations: 2017Sa70, 2004Pf02, 2003Gr04, 2005Pf01, 2003Gr24, 2001Gr29.
Other structure calculations: 2005Wa15, 2005Zh28, 2002Zh09, 2001Yo12, 1998Co30, 1997Co19, 1997Pa38, 1996Sh20, 1995La03.
Giant-resonance calculations: 2021Li26, 2001Sa57, 1997Ha57, 2003Ma71, 2001Sa77, 1999Sa22, 1998Ha09, 1998Ha39, 1998Sa28, 1997Ha55.

^{34}Ca Levels

E(level)	J^{π}	$T_{1/2}$	Comments
0?	0^{+}	<35 ns	%p=?; %2p=? $T_{1/2}$: upper limit estimated from expected cross sections of a ^{40}Ca beam fragmentation on a Ni target at LISE, GANIL (1996PoZZ). The authors of 1986La17 provided additional data from their experiment in a private communication 1996PoZZ.